

RODRIGO NOVELO

SOFTWARE ENGINEER

CONTACT

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Redondo, Évora

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in/rodrigo-novelo-997079332

EDUCATION

ISEC | Coimbra | 2021-2025

B.Sc. Computer Engineering

Networks and Systems Administration

CERTIFICATIONS

Cisco: CCNA: Introduction to Networks.

Cisco: CCNA: Network Security.

TryHackMe: Cyber Security 101.

SOFT SKILLS

- Adaptability
- Critical Thinking & Problem Solving
- Team Collaboration
- Time Management
- Continuos Learning

LANGUAGES

- Portuguese: Native
- English: C2 (Proficient)
- Spanish: B1 (Intermediate)

PROFILE

B.Sc. in Computer Engineering with a focus on Frontend Development, AI Systems, and Network Security. Strong foundation in Object-Oriented Programming (OOP) and Data Structures & Algorithms, with hands-on experience building enterprise web applications using Angular and TypeScript. Published work in LLM automation.

WORK EXPERIENCE

Closer Consulting | Lisboa
Frontend Developer
APR 2026 - PRESENT
[6 MONTHS]

STACK:

Angular, TypeScript, RxJS, SCSS, PrimeNG, SPMS Design System, OIDC/JWT, Power BI Embedded, STOMP/WebSockets.

- Built enterprise-grade features using modern framework capabilities (Standalone Components, Signals) to deliver seamless, responsive experiences across mobile and desktop.
- Deliver end-to-end features including reactive forms, dynamic routing and reusable UI directives.
- Integrate Swagger/OpenAPI REST endpoints (auth, pagination, file exports) and WebSocket feeds; report backend contract gaps.
- Fix bugs and improve app performance by resolving memory leaks and UI permission issues.
- Use Azure DevOps for deployments, actively reviewing code (Pull Requests) and testing new features.

KEY PROJECTS

AI Email Assistant
Full-Stack Project | @Isec
MAR 2025 - JUL 2025

- Engineered a RAG-based backend using Python and Google Gemini for context-aware automation.
- Implemented a relational database schema using SQLAlchemy and SQLite for robust data persistence.
- Developed an asynchronous architecture using Celery and Redis to manage high-concurrency tasks.
- Built a Web Dashboard for manual review, integrating secure OAuth 2.0 authentication.
- Optimized data retrieval through a hybrid search engine using Sentence Transformers.

Enterprise Healthcare Web Platform
@Closer
APR 2026 - PRESENT

- Developed responsive Angular modules (Standalone Components, Signals) and integrated OpenAPI/WebSocket services.



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Transformer & Backpropagation Engine

- Built neural network architectures from the ground up in PyTorch, implementing backpropagation, gradient descent, and multi-head self-attention without high-level abstractions to understand the mechanics behind modern transformer models. Wrote custom tokenization and data-preprocessing pipelines to train compact language models locally, covering the full loop from raw text to a converged checkpoint.

Self-Hosted LLM Platform

- Served a quantized 8B model locally behind a streaming API with caching, auth, and PII redaction.
- Fine-tuned it with LoRA on a single GPU, validating the improvement on a held-out benchmark.
- Built a regression-testing pipeline that blocks prompt or model changes that lower output quality.

GitOps Home Server

- Defined a full self-hosted server as code (Docker Compose + Ansible), rebuildable from a blank OS.
- Built CI/CD that deploys on every push, with health checks and automatic rollback on failure.
- Added encrypted backups with automated restore tests, and a React dashboard for service status and resource usage.

TECHNICAL SKILLS

Programming Languages
TypeScript, JavaScript, Python, C/C++, Bash.

Frontend
Angular (5–20), React, HTML5, Sass, PrimeNG, Tailwind CSS; NgRx, NgXs, RxJS.

Backend & APIs
REST (OpenAPI/Swagger), WebSockets, OAuth 2.0, Celery, Redis.

Databases
PostgreSQL, MySQL, SQLite.

AI & ML
PyTorch, Hugging Face Transformers, PEFT/LoRA, RAG, LangChain.

DevOps and Infra
Git, Azure DevOps, CI/CD, Docker, Ansible, Nginx, Linux (Ubuntu/Debian/Kali).



PUBLICATIONS

2025: Novelo, R., Silva, R. R., & Bernardino, J. "A Literature Review of Personalized Large Language Models for Email Generation and Automation." Future Internet, 17(12), 536. MDPI.

Link: https://www.mdpi.com/1999-5903/17/12/536 | DOI: 10.3390/fi17120536.



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ACADEMIC BACKGROUND

Year 1

- Calculus I
- Electronics
- Introduction to Programming
- Digital Systems
- Web Technologies
- Linear Algebra
- Calculus II
- Computer Graphics Fundamentals
- Scripting Languages
- Statistical Methods
- Programming
- Computer Architecture

Year 2

- Databases
- Introduction to Artificial Intelligence
- Introduction to Computer Networks
- Operations Research
- Object-Oriented Programming
- Operating Systems
- Structured Cabling
- Data Routing
- Management
- Network & Information Security
- Network Services I
- Operating Systems II

Year 3

- Availability & Performance
- Network Management
- Web Programming
- Network Services II
- Link/Switching Technologies
- Professional Ethics
- Final Year Project: AI Email Assistant - RAG-based email automation; resulted in a peer-reviewed publication (Future Internet, MDPI, 2025).